You may create the layout of the full-adder by hand in order to create a compact design. Notice that the AND/OR combination of cells may be replaced by a complex gate. Alternatively, you may use DSCH2 to create the schematic diagram of the full-adder and compile it directly into layout. In the compiled version of the adder shown in figure 6-9, the XOR gate is routed on the left and the AND gate is routed on the right. Click on **Simulate→Start Simulation** and verify the truth table of the full-adder.

6.7 Four-Bit Adder

The four-bit adder circuit includes adders in serial to perform the arithmetic addition. The result of each stage propagates to the next one, from the right to left (Figure 6-10).

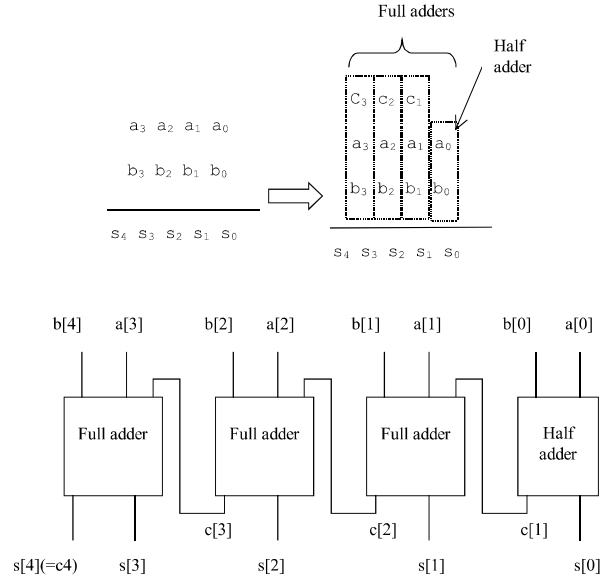


Fig. 6-10. Structure of the 4-bit ripple-carry adder

The circuit of figure 6-11 allows a four-bit addition between two numbers A3,A2,A1,A0 and B3,B2,B1,B0. Insert the user-defined `Fadd.sym` symbol using the command **Insert → User Symbol**. In DSCH2, the A and B numbers are generated by keyboard symbols, as reported below. Also notice the hexadecimal displays.

The two displays are connected to the identical data, but are configured in different mode: hexadecimal format for the right-most display, and integer mode for the left-most display. To change the display mode, double click inside the symbol, and change the format in the symbol property window. The default option is the hexadecimal format. The unsigned integer format is used in the schematic diagram of figure 6-11 for the left display. Integer and fixed point display formats are used for arithmetic circuits.

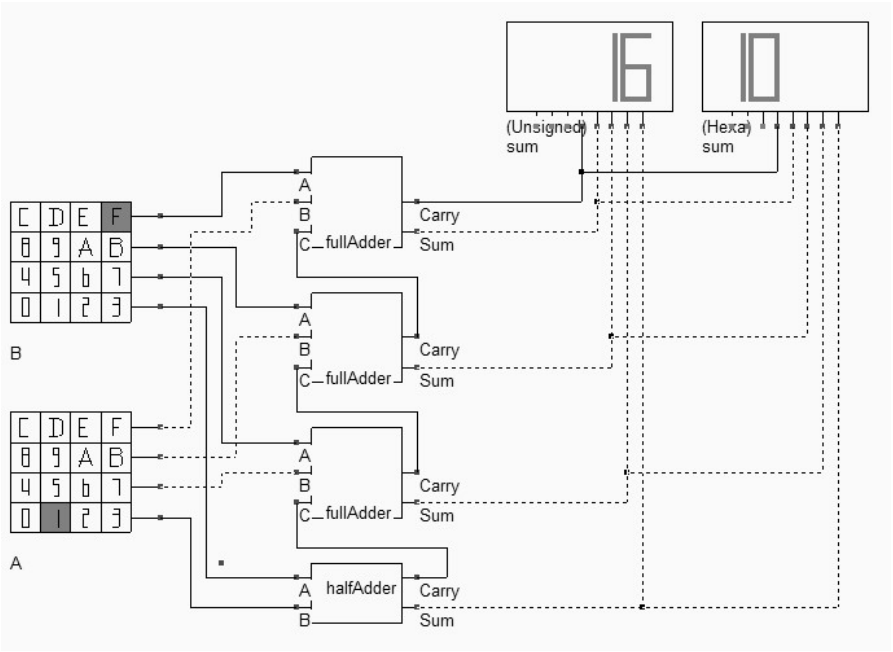


Fig. 6-11. Schematic diagram of the four-bit adder and some examples of results (Add4.SCH).

Figure 6-12 details the four-bit adder layout based on the full-custom cell design, with the corresponding simulation. In Microwind2, the command **Edit → Duplicate X,Y** has been used to duplicate the full-adder layout vertically.

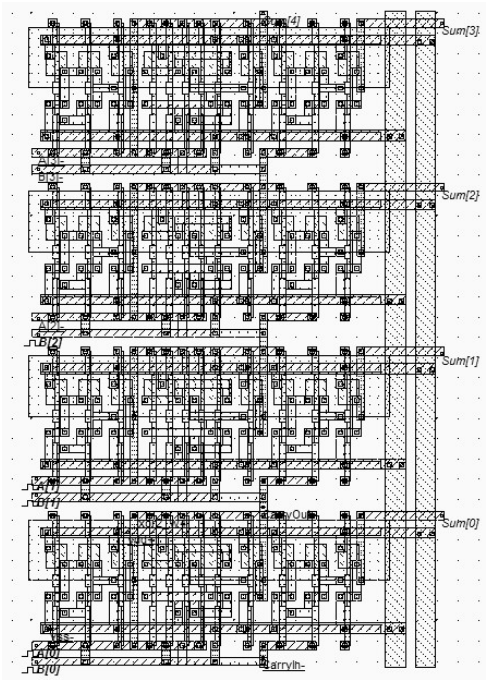


Fig. 6-12. Manual design of the four-bit adder (ADD4.MSK).

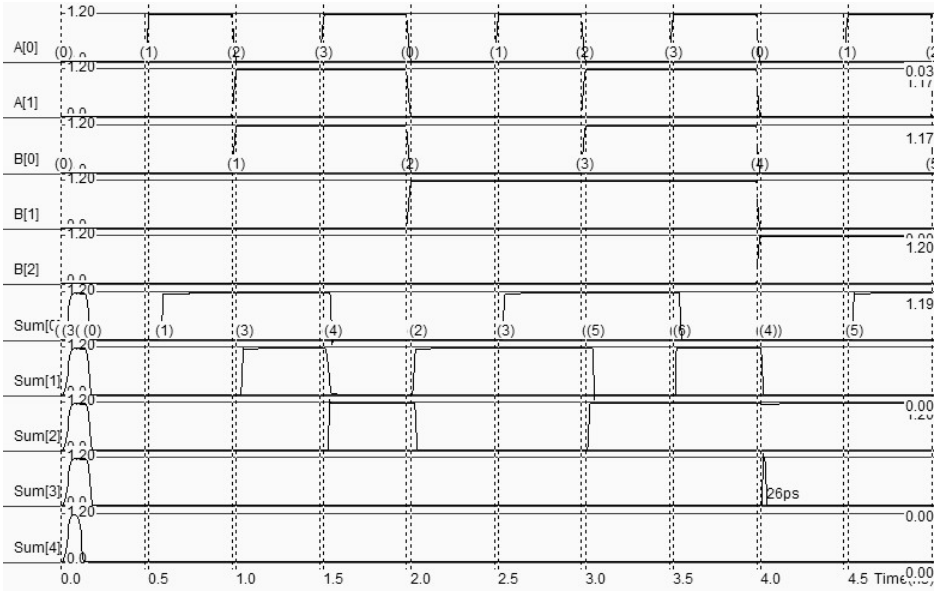


Fig. 6-13. Simulation of the four-bit adder (ADD4.MSK).

6.8 Comparator

The truth table and the schematic diagram of the comparator are given below. The A=B equality represents an XNOR gate, and A>B, A<B are operators obtained by using inverters and AND gates.

Comparator

A	B	A>B	A<B	A=B
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

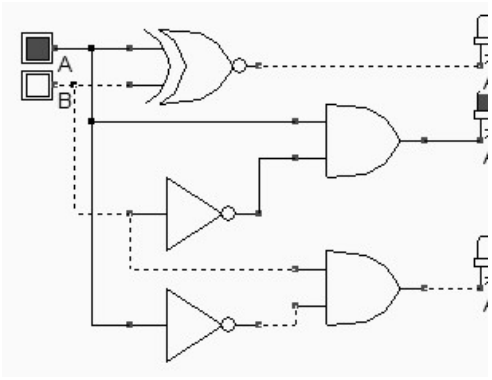


Fig. 6-14 The truth table and schematic diagram of the comparator (COMP.SCH).

Using DSCH2, the logic circuit of the comparator is designed and verified at logic level. Then the conversion into Verilog is invoked (**File → Make verilog File**). Microwind2 compiles the verilog text into layout. The simulation of the comparator is given in Figure 6-15. The XNOR gate is located at the left side of the design. The inverter and NOR gates are at the right side. After the initialization, *A=B* rises to 1. The clocks *A* and *B* produce the combinations 00,01,10 and 11.